HW #6: WRITTEN

Student Name _____

Box answers and show all work in the space provided. Submit at the start of class on the scheduled due date.

For the following questions, consider the manipulator shown in Figure 1. The first axis is a rotation about the z-axis of the spatial frame. The second axis is a rotation about the y-axis of the spatial frame as shown.

1) Fine each joint axis for FK calculations.
   Use ω for rotations and v for translations.

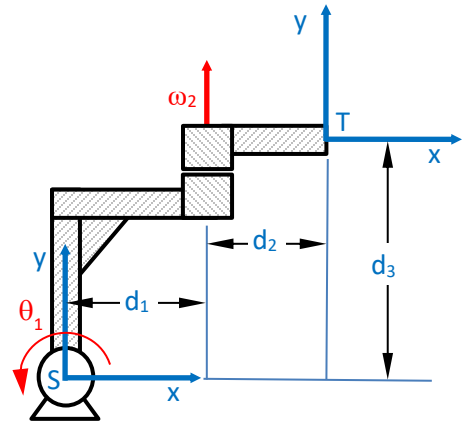

Figure 1. A 2 DOF manipulator shown in the home configuration.

2) Find a point on each joint axis, $\mathbf{q}_{1-2}$. Indicated not-needed as necessary.

3) Find the twists, ξ_{1-2}, necessary for kinematic analysis.

4) Find the home position, $\mathbf{g}_{st}(0)$, of the tool frame.

5) Find the spatial manipulator Jacobian, $\mathbf{J}_{st}^{s}(\theta)$. You can confirm via Matlab but please show the calculations steps.

$$e^{\hat{\xi}_1\theta_1} = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$e^{\hat{\xi}_2\theta_2} = \begin{bmatrix} c_2 & 0 & s_2 & d_1(1-c_2) \\ 0 & 1 & 0 & 0 \\ -s_2 & 0 & c_2 & d_1 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{p}_{st}(\theta) = \begin{bmatrix} d_1 c_1 - d_3 s_1 + d_2 c_1 c_2 \\ d_3 c_1 + d_1 s_1 + d_2 c_2 s_1 \\ -d_2 s_2 \end{bmatrix}$$